

1. Create MariaDB DB on EC2.

Implementation

- Launch an EC2 instance with **Amazon Linux 2023**.
- Connect to the EC2 instance using SSH.
- Update the packages: `sudo dnf update -y`
- Install MariaDB: `sudo dnf install mariadb1011-server -y`
- Start MariaDB: `sudo systemctl start mariadb`
- Enable MariaDB to start automatically when the server boots:
`sudo systemctl enable mariadb`
- Verify the MariaDB service: `sudo systemctl status mariadb`
- Login to MariaDB: `sudo mariadb`
- Create the database: `CREATE DATABASE companydb;`
- Select the database: `USE companydb;`
- Create the employees table:
- `CREATE TABLE employees (`
- `id INT PRIMARY KEY AUTO_INCREMENT,`
- `name VARCHAR(100),`
- `department VARCHAR(100),`
- `salary INT`
- `);`
- Insert the initial employee data:
- `INSERT INTO employees (name, department, salary) VALUES`
- `('Ravi', 'DevOps', 60000),`
- `('Poojith', 'Cloud', 65000),`
- `('Rahul', 'Development', 55000),`
- `('Anil', 'Testing', 50000),`
- `('Suresh', 'DevOps', 70000);`
- Verify the data:
- `SELECT * FROM employees;`
- Confirm that the companydb database and employees table were created successfully.

```
[ec2-user@ip-10-0-1-140 ~]$ sudo dnf install mariadb1011-server -y
Last metadata expiration check: 0:00:45 ago on Fri Sep 25 06:35:29 2026.
Dependencies resolved.
=====
Package                               Architecture      Version                               Repository      Size
-----
Installing:
mariadb1011-server                   x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     11 M
Installing dependencies:
lzo                                   x86_64            2.10-4.amzn2023.0.2                 amazonlinux     76 k
mariadb-connector-c                   x86_64            3.3.10-1.amzn2023.0.2               amazonlinux     211 k
mariadb-connector-c-config            noarch            3:3.10-1.amzn2023.0.2               amazonlinux     9.3 k
mariadb1011                           x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     1.8 M
mariadb1011-common                   x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     30 k
mariadb1011-errmsg                     x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     256 k
mysql-selinux                          noarch            1.0.14-2.amzn2023.0.1               amazonlinux     36 k
perl-DBD-MariaDB                      x86_64            1.22-1.amzn2023.0.5                 amazonlinux     152 k
perl-DBI                              x86_64            1.652-1.amzn2023.0.1               amazonlinux     811 k
perl-File-Copy                         noarch            2.34-477.amzn2023.0.9               amazonlinux     20 k
perl-Math-BigInt                       noarch            1.1.1.9998.39-2.amzn2023.0.2        amazonlinux     202 k
perl-Math-BigIntat                     noarch            0.2624-500.amzn2023.0.2             amazonlinux     42 k
perl-Math-Complex                      noarch            1.59-477.amzn2023.0.9               amazonlinux     47 k
perl-Module-Load                      noarch            1.0.36-2.amzn2023.0.2               amazonlinux     18 k
perl-Geo-HashName                     x86_64            1.24-477.amzn2023.0.9               amazonlinux     17 k
snappy                                 x86_64            1.1.8-5.amzn2023.0.2               amazonlinux     35 k
Installing weak dependencies:
mariadb1011-backup                    x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     6.6 M
mariadb1011-client-utils              x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     40 k
mariadb1011-cracklib-password-check   x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     17 k
mariadb1011-ssapi-server              x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     18 k
mariadb1011-server-utils              x86_64            3:10.11.18-1.amzn2023.0.1           amazonlinux     269 k
=====
Transaction Summary
Install 22 Packages
Total download size: 22 M
Installed size: 143 M
Downloading Packages:
(1/22): lzo-2.10-4.amzn2023.0.2.x86_64.rpm                                3.1 MB/s | 76 kB  00:00
(2/22): mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch.rpm       322 kB/s | 9.3 kB 00:00
(3/22): mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64.rpm              6.3 MB/s | 211 kB 00:00
(4/22): mariadb1011-10.11.18-1.amzn2023.0.1.x86_64.rpm                  46 MB/s | 1.8 MB 00:00
(5/22): mariadb1011-client-utils-10.11.18-1.amzn2023.0.1.x86_64.rpm       1.1 MB/s | 40 kB 00:00
(6/22): mariadb1011-cracklib-password-check-10.11.18-1.amzn2023.0.1.x86_64.rpm 640 kB/s | 17 kB 00:00
(7/22): mariadb1011-common-10.11.18-1.amzn2023.0.1.x86_64.rpm            981 kB/s | 30 kB 00:00
(8/22): mariadb1011-backup-10.11.18-1.amzn2023.0.1.x86_64.rpm            64 MB/s | 6.6 MB 00:00
(9/22): mariadb1011-ssapi-server-10.11.18-1.amzn2023.0.1.x86_64.rpm       339 kB/s | 18 kB 00:00
(10/22): mariadb1011-errmsg-10.11.18-1.amzn2023.0.1.x86_64.rpm           6.4 MB/s | 256 kB 00:00
(11/22): mariadb1011-server-utils-10.11.18-1.amzn2023.0.1.x86_64.rpm       11 MB/s | 269 kB 00:00
(12/22): mysql-selinux-1.0.14-2.amzn2023.0.1.noarch.rpm                  1.2 MB/s | 36 kB 00:00
(13/22): perl-DBD-MariaDB-1.22-1.amzn2023.0.5.x86_64.rpm                 6.7 MB/s | 152 kB 00:00
(14/22): perl-DBI-1.652-1.amzn2023.0.1.x86_64.rpm                        22 MB/s | 811 kB 00:00
(15/22): perl-File-Copy-2.34-477.amzn2023.0.9.noarch.rpm                 867 kB/s | 20 kB 00:00
(16/22): perl-Math-BigInt-1.9998.39-2.amzn2023.0.2.noarch.rpm            8.0 MB/s | 202 kB 00:00
(17/22): perl-Math-BigIntat-0.2624-500.amzn2023.0.2.noarch.rpm           1.5 MB/s | 42 kB 00:00
(18/22): perl-Math-Complex-1.59-477.amzn2023.0.9.noarch.rpm              1.8 MB/s | 47 kB 00:00
(19/22): perl-Module-Load-1.0.36-2.amzn2023.0.2.noarch.rpm               674 kB/s | 18 kB 00:00
=====
```

```
[ec2-user@ip-10-0-1-140 ~]$ sudo systemctl start mariadb
[ec2-user@ip-10-0-1-140 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-10-0-1-140 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-09-25 06:38:32 UTC; 17s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
  Main PID: 27132 (mariabdb)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 1059)
    Memory: 81.6M
       CPU: 667ms
    CGroup: /system.slice/mariadb.service
            └─27132 /usr/libexec/mariabdb --basedir=/usr

Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: The second is mysql@localhost, it has no password either, but
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: you need to be the system 'mysql' user to connect.
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: After connecting you can set the password, if you would need to be
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: able to connect as any of these users with a password and without sudo
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: See the MariaDB Knowledgebase at https://mariadb.com/kb
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: Please report any problems at https://mariadb.org/jira
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: The latest information about MariaDB is available at https://mariadb.org/.
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: Consider joining MariaDB's strong and vibrant community:
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal mariadb-prepare-db-dir[27087]: https://mariadb.org/get-involved/
Sep 25 06:38:32 ip-10-0-1-140.ec2.internal systemd[1]: Started mariadb.service - MariaDB 10.11 database server.
```

```
[ec2-user@ip-10-0-1-140 ~]$ mariadb --version
mariadb Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using Editline wrapper
[ec2-user@ip-10-0-1-140 ~]$ sudo mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE companydb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| companydb |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.000 sec)
```

2. Insert some dummy data.

Implementation:

- Login to MariaDB: `sudo mariadb`
- Select the database: `USE companydb;`
- Insert sample employee records:

- INSERT INTO employees (name, department, salary) VALUES
- ('Ravi', 'DevOps', 60000),
- ('Poojith', 'Cloud', 65000),
- ('Rahul', 'Development', 55000),
- ('Anil', 'Testing', 50000),
- ('Suresh', 'DevOps', 70000);
- Verify the inserted records: SELECT * FROM employees;
- Verify the number of records: SELECT COUNT(*) FROM employees;
- Confirm that the dummy data is available in the employees table.

```
MariaDB [(none)]> USE companydb;
Database changed
MariaDB [companydb]> CREATE TABLE employees (
  -> id INT PRIMARY KEY AUTO_INCREMENT,
  -> name VARCHAR(100),
  -> department VARCHAR(100),
  -> salary INT
  -> );
Query OK, 0 rows affected (0.012 sec)

MariaDB [companydb]> SHOW TABLES;
+-----+
| Tables_in_companydb |
+-----+
| employees            |
+-----+
1 row in set (0.000 sec)

MariaDB [companydb]> INSERT INTO employees (name, department, salary)
  -> VALUES,
  -> ('Ravi', 'DevOps', 60000),
  -> ('Poojith', 'Cloud', 65000),
  -> ('Rahul', 'Development', 55000),
  -> ('Anil', 'Testing', 50000),
  -> ('Suresh', 'DevOps', 70000);
Query OK, 5 rows affected (0.002 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [companydb]> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name  | department | salary |
+----+-----+-----+-----+
| 1  | Ravi  | DevOps    | 60000  |
| 2  | Poojith | Cloud     | 65000  |
| 3  | Rahul | Development | 55000  |
| 4  | Anil  | Testing   | 50000  |
| 5  | Suresh | DevOps    | 70000  |
+----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [companydb]> SELECT COUNT(*) FROM employees;
+-----+
| COUNT(*) |
+-----+
| 5        |
+-----+
1 row in set (0.000 sec)
```

3. Take the backup of dummy data on EC2.

Implementation

- Exit from the MariaDB client if required: EXIT;
- Create a backup of the companydb database:
 - sudo mariadb-dump companydb > companydb_backup.sql
- Verify that the backup file was created: ls -lh companydb_backup.sql
- Check the contents of the backup file: head companydb_backup.sql
- Verify that the SQL file contains the database structure and employee data.
- The backup file **companydb_backup.sql** was used later to migrate the database from EC2 to Amazon RDS.

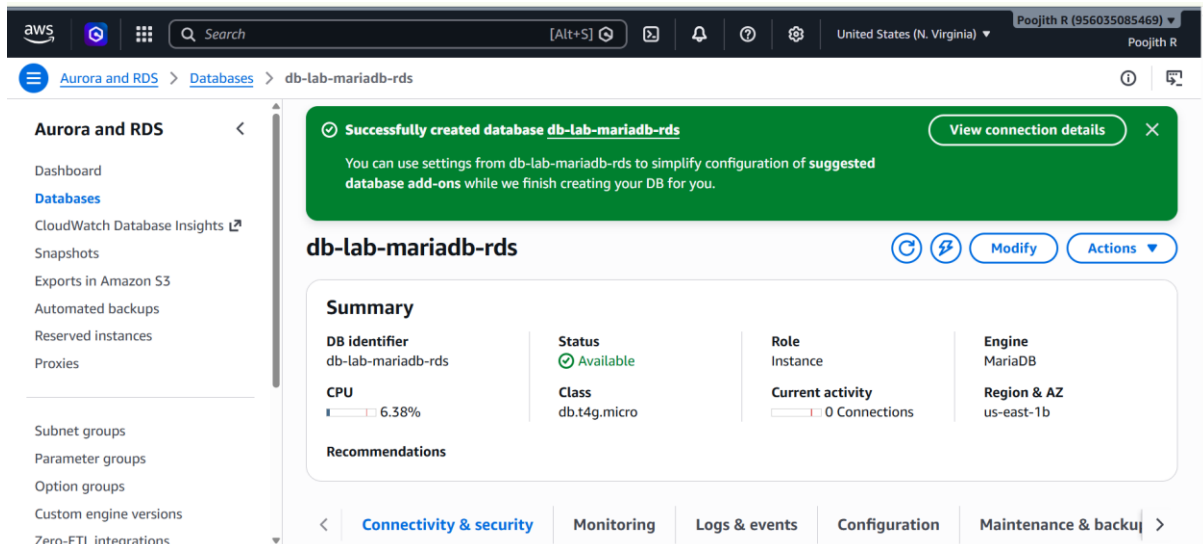
```
[ec2-user@ip-10-0-1-140 ~]$ sudo mariadb-dump companydb > companydb_backup.sql
[ec2-user@ip-10-0-1-140 ~]$ ls -lh companydb_backup.sql
-rw-r--r--. 1 ec2-user ec2-user 2.2K Sep 25 06:46 companydb_backup.sql
[ec2-user@ip-10-0-1-140 ~]$ head -30 companydb_backup.sql
/*M19999999\ enable the sandbox mode */
-- MariaDB dump 10.19 Distrib 10.11.18-MariaDB, for Linux (x86_64)
--
-- Host: localhost Database: companydb
--
-- Server version 10.11.18-MariaDB
--
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
--
-- Table structure for table `employees`
--
DROP TABLE IF EXISTS `employees`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8mb4 */;
CREATE TABLE `employees` (
  `id` int(11) NOT NULL AUTO_INCREMENT,
  `name` varchar(100) DEFAULT NULL,
  `department` varchar(100) DEFAULT NULL,
  `salary` int(11) DEFAULT NULL,
[ec2-user@ip-10-0-1-140 ~]$ grep -n "Poojith" companydb_backup.sql
43:(2,'Poojith','Cloud',65000),
[ec2-user@ip-10-0-1-140 ~]$ grep -n "Ravi" companydb_backup.sql
42:(1,'Ravi','DevOps',60000),
[ec2-user@ip-10-0-1-140 ~]$
```

4. Launch MariaDB RDS instance.

Implementation

- Open **AWS Console** → **RDS**.
- Select **Create database**.
- Select **Standard create**.
- Select **MariaDB** as the database engine.
- Select the required MariaDB version.
- Select the required database template.
- Enter the DB instance identifier: db-lab-mariadb-rds
- Configure the master username: admin
- Configure the master password.
- Select the required DB instance class.
- Configure the required storage.
- Select the VPC: Database-Lab-VPC
- Select the DB subnet group: db-lab-private-subnet-group
- Configure the RDS security group: DB-Lab-MariaDB-RDS-SG
- Allow **TCP port 3306** from the EC2 security group.
- Configure the RDS instance with **Public access = No**.
- Click **Create database**.
- Wait until the RDS instance status becomes:
- Available

- Copy the RDS endpoint: db-lab-mariadb-rds.c672saeaiaul.us-east-1.rds.amazonaws.com
- The RDS endpoint was used to connect to the MariaDB database from the EC2 instance.



5. Migrate database from EC2 to RDS.

Implementation

- Ensure that the EC2 security group can communicate with the RDS security group on **TCP port 3306**.
- From the MariaDB EC2 instance, test the connection to RDS: `mariadb -h db-lab-mariadb-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p`
- After connecting to RDS, create the target database:
`CREATE DATABASE companydb;`
- Exit from the MariaDB client: `EXIT;`
- Import the EC2 database backup into RDS:
`mariadb -h db-lab-mariadb-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p companydb < companydb_backup.sql`
- Login again to the RDS MariaDB database:
`mariadb -h db-lab-mariadb-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p`
- Select the migrated database: `USE companydb;`
- Verify the tables: `SHOW TABLES;`
- Verify the migrated employee data: `SELECT * FROM employees;`
- Confirm that all 5 employee records from the EC2 database are available on RDS.

```
[ec2-user@ip-10-0-1-140 ~]$ mariadb -h db-lab-mariadb-rds.c672saeiaul.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 44
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE companydb;
Query OK, 1 row affected (0.009 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| companydb |
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.003 sec)

MariaDB [(none)]> EXIT;
Bye
[ec2-user@ip-10-0-1-140 ~]$ mariadb -h db-lab-mariadb-rds.c672saeiaul.us-east-1.rds.amazonaws.com -u admin -p companydb < companydb_backup.sql
Enter password:
[ec2-user@ip-10-0-1-140 ~]$ mariadb -h db-lab-mariadb-rds.c672saeiaul.us-east-1.rds.amazonaws.com -u admin -p companydb
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 48
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
MariaDB [companydb]> SHOW TABLES;
+-----+
| Tables_in_companydb |
+-----+
| employees |
+-----+
1 row in set (0.001 sec)

MariaDB [companydb]> SELECT * FROM employees;
+----+-----+-----+-----+
| id | name  | department | salary |
+----+-----+-----+-----+
| 1  | Ravi  | DevOps    | 60000  |
| 2  | Poojith | Cloud    | 65000  |
| 3  | Rahul | Development | 55000  |
| 4  | Anil  | Testing   | 50000  |
| 5  | Suresh | DevOps    | 70000  |
+----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [companydb]> SELECT COUNT(*) FROM employees;
+-----+
| COUNT(*) |
+-----+
| 5 |
+-----+
1 row in set (0.001 sec)

MariaDB [companydb]> EXIT;
Bye
[ec2-user@ip-10-0-1-140 ~]$
```

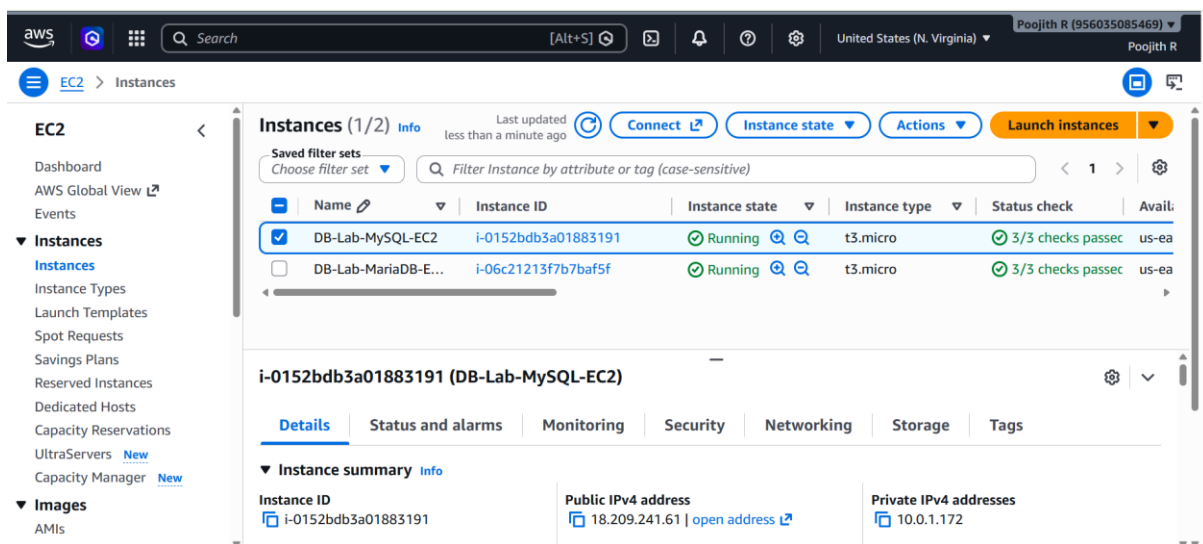
6. Install MySQL DB on EC2.

Implementation

- Launch a separate EC2 instance with **Amazon Linux 2023**.
- Connect to the EC2 instance using SSH.
- Install MySQL Community Server.
- Start the MySQL service: `sudo systemctl start mysqld`
- Enable MySQL to start automatically: `sudo systemctl enable mysqld`
- Verify the MySQL service: `sudo systemctl status mysqld`
- Verify the installed MySQL version: `mysql --version`
- Login to MySQL: `mysql -u root -p`
- Create the database: `CREATE DATABASE salesdb;`
- Select the database: `USE salesdb;`
- Create the orders table:


```
CREATE TABLE orders (
  id INT PRIMARY KEY AUTO_INCREMENT,
  customer_name VARCHAR(100),
  product VARCHAR(100),
  amount DECIMAL(10,2)
);
```

- Insert sample order data:
 INSERT INTO orders (customer_name, product, amount) VALUES
 ('Ravi', 'Laptop', 75000),
 ('Poojith', 'Monitor', 25000),
 ('Rahul', 'Keyboard', 5000),
 ('Anil', 'Mouse', 2000),
 ('Suresh', 'Headset', 3500);
- Verify the data: SELECT * FROM orders;
- Verify the number of records: SELECT COUNT(*) FROM orders;
- Confirm that the MySQL database and sample data were created successfully.



```
[ec2-user@ip-10-0-1-172 ~]$ sudo dnf install https://dev.mysql.com/get/mysql84-community-release-el9-1.noarch.rpm -y
Last metadata expiration check: 0:00:35 ago on Fri Sep 23 07:25:05 2026.
mysql84-community-release-el9-1.noarch.rpm
Dependencies resolved.
106 kB/s | 13 kB | 00:00
-----
Package                               Architecture Version      Repository      Size
Installing:
mysql84-community-release             noarch       el9-1         @commandline   13 k
Transaction Summary
-----
Install 1 Package
Total size: 13 k
Installed size: 14 k
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : mysql84-community-release-el9-1.noarch
  Installing     : mysql84-community-release-el9-1.noarch
  Verifying      : mysql84-community-release-el9-1.noarch
Installed:
mysql84-community-release-el9-1.noarch
complete!
[ec2-user@ip-10-0-1-172 ~]$ sudo dnf repolist enabled | grep -i mysql
mysql-8.4-lts-community               MySQL 8.4 LTS Community Server
mysql-connectors-community            MySQL Connectors Community
mysql-tools-8.4-lts-community         MySQL Tools 8.4 LTS Community
[ec2-user@ip-10-0-1-172 ~]$ sudo dnf list available | grep -E 'mysql-community-server'
mysql-community-server.x86_64        8.4.11-1.el9      mysql-8.4-lts-community
mysql-community-server-debug.x86_64 8.4.11-1.el9      mysql-8.4-lts-community
[ec2-user@ip-10-0-1-172 ~]$ sudo dnf install mysql-community-server -y
Last metadata expiration check: 0:00:16 ago on Fri Sep 23 07:26:04 2026.
Dependencies resolved.
-----
Package                               Architecture Version      Repository      Size
Installing:
mysql-community-server                x86_64       8.4.11-1.el9  mysql-8.4-lts-community 52 M
Installing dependencies:
mysql-community-client                x86_64       8.4.11-1.el9  mysql-8.4-lts-community 3.4 M
mysql-community-client-plugins        x86_64       8.4.11-1.el9  mysql-8.4-lts-community 1.5 M
mysql-community-common                x86_64       8.4.11-1.el9  mysql-8.4-lts-community 581 k
mysql-community-icu-data-files        x86_64       8.4.11-1.el9  mysql-8.4-lts-community 2.3 M
mysql-community-liba                  x86_64       8.4.11-1.el9  mysql-8.4-lts-community 1.5 M
Transaction Summary
-----
Install 6 Packages
Total download size: 62 M
```

```

complete!
[ec2-user@ip-10-0-1-172 ~]$ sudo systemctl start mysqld
[ec2-user@ip-10-0-1-172 ~]$ sudo systemctl enable mysqld
[ec2-user@ip-10-0-1-172 ~]$ sudo systemctl status mysqld
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-09-25 07:26:47 UTC; 15s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
  Main PID: 26246 (mysqld)
    Status: "Server is operational"
     Tasks: 35 (limit: 1059)
    Memory: 451.0M
       CPU: 4.382s
   CGroup: /system.slice/mysqld.service
           └─26246 /usr/sbin/mysqld

Sep 25 07:26:38 ip-10-0-1-172.ec2.internal systemd[1]: Starting mysqld.service - MySQL Server...
Sep 25 07:26:47 ip-10-0-1-172.ec2.internal systemd[1]: Started mysqld.service - MySQL Server.
[ec2-user@ip-10-0-1-172 ~]$ mysql --version
mysql Ver 8.4.11 for Linux on x86_64 (MySQL Community Server - GPL)
[ec2-user@ip-10-0-1-172 ~]$ sudo grep 'temporary password' /var/log/mysqld.log
2026-09-25T07:26:43.433218Z 6 [Note] [MV-010454] [Server] A temporary password is generated for root@localhost: ecws_2Zln_27
[ec2-user@ip-10-0-1-172 ~]$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> EXIT;
Bye
[ec2-user@ip-10-0-1-172 ~]$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.4.11

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```

```

[ec2-user@ip-10-0-1-172 ~]$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.4.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE salesdb;
Query OK, 1 row affected (0.01 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| salesdb |
| sys |
+-----+
5 rows in set (0.01 sec)

mysql> USE salesdb;
Database changed
mysql> CREATE TABLE orders (
  ->   id INT PRIMARY KEY AUTO INCREMENT,
  ->   customer_name VARCHAR(100),
  ->   product VARCHAR(100),
  ->   amount DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO orders (customer_name, product, amount)
  -> VALUES
  -> ('Ravi', 'Laptop', 75000.00),
  -> ('Poojith', 'Monitor', 25000.00),
  -> ('Rahul', 'Keyboard', 5000.00),
  -> ('Anil', 'Mouse', 2000.00),
  -> ('Suresh', 'Headset', 3500.00);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

mysql> INSERT INTO orders (customer_name, product, amount)
  -> VALUES
  -> ('Ravi', 'Laptop', 75000.00),
  -> ('Poojith', 'Monitor', 25000.00),
  -> ('Rahul', 'Keyboard', 5000.00),
  -> ('Anil', 'Mouse', 2000.00),
  -> ('Suresh', 'Headset', 3500.00);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM orders;
+----+ customer_name | product | amount |
+----+-----+
| 1 | Ravi | Laptop | 75000.00 |
| 2 | Poojith | Monitor | 25000.00 |
| 3 | Rahul | Keyboard | 5000.00 |
| 4 | Anil | Mouse | 2000.00 |
| 5 | Suresh | Headset | 3500.00 |
+----+-----+
5 rows in set (0.00 sec)

mysql> SELECT COUNT(*) FROM orders;
+-----+
| COUNT(*) |
+-----+
| 5 |
+-----+
1 row in set (0.01 sec)

mysql> EXIT;
Bye
[ec2-user@ip-10-0-1-172 ~]$

```

7. Launch MySQL RDS image.

- Open **AWS Console** → **RDS**.
- Click **Create database**.
- Select **Standard create**.
- Wait until the RDS instance status becomes: Available

- Copy the RDS endpoint: db-lab-mysql-rds.c672saeaiaul.us-east-1.rds.amazonaws.com

Connect from EC2

- From the MySQL EC2 instance, test the RDS connection:
- `mysql -h db-lab-mysql-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p`
- Create the target database: `CREATE DATABASE salesdb;`
- Exit from the RDS MySQL client: `EXIT;`

Create Backup

- On the MySQL EC2 instance, create a backup:
`mysqldump -u root -p salesdb > salesdb_backup.sql`
- Verify the backup file: `ls -lh salesdb_backup.sql`
- Import the backup into MySQL RDS:
`mysql -h db-lab-mysql-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p salesdb < salesdb_backup.sql`
- Connect to the RDS database again:
`mysql -h db-lab-mysql-rds.c672saeaiaul.us-east-1.rds.amazonaws.com -u admin -p`
- Select the database: `USE salesdb;`
- Verify the table: `SHOW TABLES;`
- Verify the migrated data: `SELECT * FROM orders;`
- Confirm that all 5 order records were successfully migrated from MySQL EC2 to MySQL RDS.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information (Poojith R, 956035085469). Below the navigation bar, the breadcrumb trail indicates the path: Aurora and RDS > Databases > db-lab-mysql-rds. On the left, a sidebar menu lists various database-related options like Dashboard, Databases, CloudWatch Database Insights, Snapshots, Exports in Amazon S3, Automated backups, Reserved instances, Proxies, Subnet groups, Parameter groups, Option groups, Custom engine versions, and Zero-ETL integrations. The main content area displays a green success message: 'Successfully created database db-lab-mysql-rds'. Below this, a summary card for the instance 'db-lab-mysql-rds' is shown, detailing its status (Available), role (Instance), engine (MySQL Community), CPU usage (4.65%), class (db.t4g.micro), current activity (0 Connections), and region (us-east-1a). At the bottom, there are tabs for 'Connectivity & security', 'Monitoring', 'Logs & events', 'Configuration', and 'Zero-ETL integrations'.

```
[ec2-user@ip-10-0-1-172 ~]$ mysql -h db-lab-mysql-rds.c672saeiaul.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.4.9 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 8.4.9     |
+-----+
1 row in set (0.00 sec)

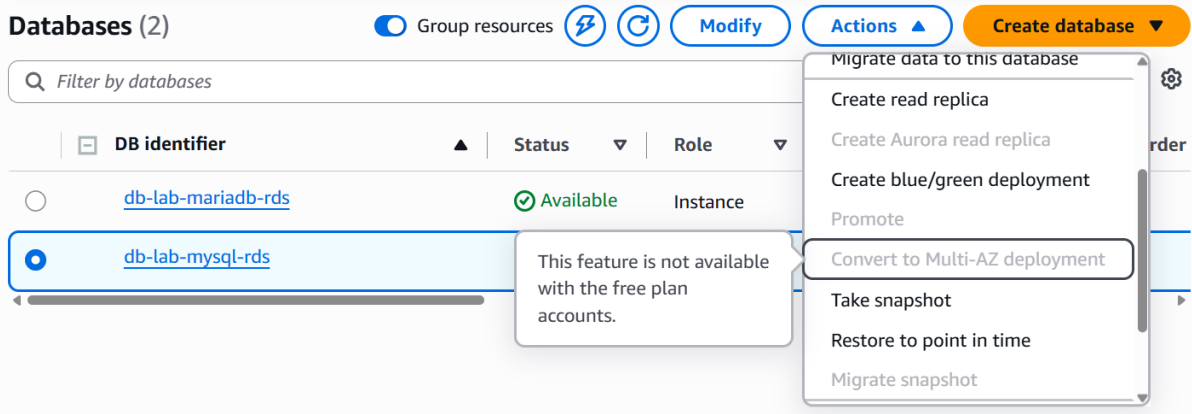
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql           |
| performance_schema |
| sys             |
+-----+
4 rows in set (0.05 sec)

mysql> EXIT;
Bye
[ec2-user@ip-10-0-1-172 ~]$
```

8. Configure Multi-AZ.

Implementation

- Open **AWS Console** → **RDS** → **Databases**.
- Select the MySQL RDS instance: db-lab-mysql-rds
- Click **Modify**.
- Open the **Availability & durability** section.
- Check the Multi-AZ deployment option.
- The DB subnet group was already configured with subnets in different Availability Zones:
 - us-east-1a
 - us-east-1b
- Attempt to enable Multi-AZ deployment.
- Select **Apply immediately** if available.
- Click **Modify DB instance**.
- The AWS account displayed a restriction indicating that this feature was not available with the Free plan account.
- Therefore, the Multi-AZ standby instance could not be created in this lab.
- Multi-AZ was studied as a **high-availability and failover** feature, but the actual configuration was not completed because of the account restriction.



9. Take backup of DB and restore the DB.

Create Database Backup

- Open **AWS Console** → **RDS** → **Databases**.
- Select the MySQL RDS instance: db-lab-mysql-rds
- Click: **Actions** → **Create snapshot**
- Wait until the snapshot status becomes: Available
- **Restore the Database**
- Go to: **RDS** → **Snapshots**
- Select the snapshot created
- **Actions** → **Restore snapshot**
- Create **Restore DB instance**.
- Wait until the restored RDS instance becomes: Available
- Copy the restored RDS endpoint.
- Connect to the restored database from the MySQL EC2 instance:
- `mysql -h <RESTORED-RDS-ENDPOINT> -u admin -p`
- Verify the databases: `SHOW DATABASES;`
- Select the database: `USE salesdb;`
- Verify the tables: `SHOW TABLES;`
- Verify the restored data: `SELECT * FROM orders;`
- Confirm that the database and its 5 records were successfully restored from the RDS snapshot.

```
poojith@Poojith WING64 ~/Downloads
$ ssh -i Test1.pem ec2-user@18.209.241.61

Amazon Linux 2023

Last login: Fri Sep 25 07:24:15 2026 from 103.203.172.230
[ec2-user@ip-10-0-1-172 ~]$ mysqldump -u root -p salesdb > salesdb_backup.sql
Enter password:
[ec2-user@ip-10-0-1-172 ~]$ ls -lh salesdb_backup.sql
-rw-r--r-- 1 ec2-user ec2-user 2.1K Sep 25 17:49 salesdb_backup.sql
[ec2-user@ip-10-0-1-172 ~]$ grep -E "CREATE TABLE|INSERT INTO" salesdb_backup.sql
CREATE TABLE orders (
INSERT INTO orders VALUES (1,'Ravi','Laptop',75000.00),(2,'Poojith','Monitor',25000.00),(3,'Rahul','Keyboard',5000.00),(4,'Anil','Mouse',2000.00),(5,'Suresh','Headset',3500.00);
[ec2-user@ip-10-0-1-172 ~]$ mysql -u root -p -e "SELECT COUNT(*) FROM salesdb.orders;"
Enter password:
+-----+
| COUNT(*) |
+-----+
|          |
+-----+
[ec2-user@ip-10-0-1-172 ~]$ mysql -h db-lab-mysql-rds.c672saeaiaw.us-east-1.rds.amazonaws.com -u admin -p -e "CREATE DATABASE salesdb;"
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'10.0.1.172' (using password: YES)
[ec2-user@ip-10-0-1-172 ~]$ mysql -h db-lab-mysql-rds.c672saeaiaw.us-east-1.rds.amazonaws.com -u admin -p -e "CREATE DATABASE salesdb;"
Enter password:
[ec2-user@ip-10-0-1-172 ~]$ mysql -h db-lab-mysql-rds.c672saeaiaw.us-east-1.rds.amazonaws.com -u admin -p salesdb < salesdb_backup.sql
Enter password:
[ec2-user@ip-10-0-1-172 ~]$ mysql -h db-lab-mysql-rds.c672saeaiaw.us-east-1.rds.amazonaws.com -u admin -p -e "SELECT * FROM salesdb.orders;"
Enter password:
+----+ customer_name | product | amount |
+----+-----+
1 | Ravi | Laptop | 75000.00 |
2 | Poojith | Monitor | 25000.00 |
3 | Rahul | Keyboard | 5000.00 |
4 | Anil | Mouse | 2000.00 |
5 | Suresh | Headset | 3500.00 |
+----+-----+
[ec2-user@ip-10-0-1-172 ~]$
```

Take DB Snapshot

Preferences
To take a DB Snapshot, choose a database and name your DB Snapshot.

DB instance
DB Instance identifier. This is the unique key that identifies a DB Instance.
db-lab-mysql-rds

Snapshot name
Identifier for the DB Snapshot.
db-lab-mysql-rds-snapshot

Snapshot identifier is case insensitive, but stored as all lower-case, as in "mysnapshot". Cannot be null, empty, or blank. Must contain from 1 to 255 alphanumeric characters or hyphens. First character must be a letter. Cannot end with a hyphen or contain two consecutive hyphens.

Cancel Take snapshot

Snapshots

Manual System Shared with me Public Backup service Exports in Amazon S3

Manual snapshots (1)

Filter by manual snapshots

Snapshot name	Engine version	DB instance or cluster
db-lab-mysql-rds-snapshot	8.4.9	db-lab-mysql-rds

10. Create read replica.

Implementation

- Open **AWS Console** → **RDS** → **Databases**.
- Select the source MySQL RDS instance.
- Click: **Actions** → **Create read replica**
- Enter a unique identifier for the read replica.
- Select the destination region.
- Select the required DB instance class.
- Select the DB subnet group.
- Configure the security group.
- Configure the required storage and other settings.
- Attempt to create the read replica.
- AWS displayed the following restriction:
You reached the maximum number of instances available with free plan accounts.

- Therefore, the read replica could not be created in this lab.

aws

Search

[Alt+S]

United States (N. Virginia)

Poojith R (956035085469)

Poojith R

Aurora and RDS

Databases

Create read replica

KMS key ID

3862923d-3004-4c7c-be22-e4426c83f162

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

✖ IAM role ARN value is invalid or does not include the required permissions for: ENHANCED_MONITORING

×

Cancel

Create read replica

aws

Search

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United States (N. Virginia)

Poojith R (956035085469)

Poojith R

Aurora and RDS

Databases

Create read replica

KMS key ID

3862923d-3004-4c7c-be22-e4426c83f162

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

✖ You reached the maximum number of instances available with free plan accounts. To remove all limitations, upgrade your account plan.

×

Cancel

Create read replica